#1 · Animal Care and Nursing

What is the normal rectal temperature range for a dog or cat?

Answer

38.0 to 39.2 degrees C. A reading above 41 without infection suggests hyperthermia and is an emergency.

#2 · Animal Care and Nursing

What is the normal resting heart rate range of an adult dog?

Answer

About 60 to 140 beats per minute. Small dogs and puppies run higher.

#3 · Animal Care and Nursing

What is the normal resting heart rate range of a cat?

Answer

About 140 to 220 beats per minute.

#4 · Animal Care and Nursing

What is the normal respiratory rate of a dog and of a cat?

Answer

Dog: 10 to 30 breaths per minute. Cat: 20 to 30 breaths per minute.

#5 · Animal Care and Nursing

What are normal TPR values for an adult horse?

Answer

Temperature 37.2 to 38.5 degrees C, heart rate 28 to 44 per minute, respiratory rate 8 to 16 per minute.

#6 · Animal Care and Nursing

What are normal TPR values for an adult cow?

Answer

Temperature 38.0 to 39.3 degrees C, heart rate 48 to 84 per minute, respiratory rate 10 to 30 per minute.

#7 · Animal Care and Nursing

What do pale mucous membranes suggest?

Answer

Anemia or poor perfusion, for example from blood loss or shock.

#8 · Animal Care and Nursing

What do icteric (yellow) mucous membranes suggest?

Answer

Liver disease or excess red cell breakdown (hemolysis) causing high bilirubin.

#9 · Animal Care and Nursing

What are the parts of a SOAP medical record entry?

Answer

Subjective (history), Objective (measurements and exam), Assessment and Plan.

#10 · Animal Care and Nursing

Why must you document every procedure and observation in the record?

Answer

The medical record is a legal document. If it is not written, it is treated as not done.

#11 · Animal Care and Nursing

How much dehydration is clinically detectable, and how do signs change at 8 percent?

Answer

Detectable at about 5 percent with tacky gums. At 8 percent there is a slow skin tent and sunken eyes.

#12 · Animal Care and Nursing

Which patients give an unreliable skin tent test for hydration?

Answer

Obese, very thin and older animals.

#13 · Animal Care and Nursing

What is the ideal score on a 9 point body condition scale?

Answer

4 to 5 out of 9. On a 5 point scale the ideal is 3.

#14 · Animal Care and Nursing

What body features show an ideal body condition in a dog?

Answer

Ribs felt with light pressure and a visible waist when viewed from above.

#15 · Animal Care and Nursing

How do you calculate resting energy requirement (RER) for a 2 to 45 kg animal?

Answer

RER in kcal per day = 30 x body weight in kg + 70. For 10 kg that is 370 kcal.

#16 · Animal Care and Nursing

What is the general formula for RER?

Answer

RER = 70 x body weight in kg raised to the power of 0.75. For 10 kg that is about 394 kcal per day.

#17 · Animal Care and Nursing

Which condition can develop in a cat that stops eating for several days?

Answer

Hepatic lipidosis, a fatty liver disease, so anorectic cats need nutrition early.

#18 · Animal Care and Nursing

When is a feeding tube used?

Answer

When the gut works but the patient will not or cannot eat enough on its own.

#19 · Animal Care and Nursing

What do you do before each tube feeding?

Answer

Check tube placement, flush with a small amount of water and warm the food to body temperature.

#20 · Animal Care and Nursing

Which behaviours suggest a dog is in pain?

Answer

Whimpering, guarding an area, panting, hiding or becoming aggressive.

#21 · Animal Care and Nursing

Which behaviours suggest a cat is in pain?

Answer

Hiding, hunching, stopping grooming, flattened ears and avoiding touch.

#22 · Animal Care and Nursing

Which pain scale is designed for cats and uses facial features?

Answer

The Feline Grimace Scale, which scores ears, orbits, muzzle, whiskers and head position.

#23 · Animal Care and Nursing

How does pain change vital signs?

Answer

It raises heart rate, respiratory rate and blood pressure.

#24 · Animal Care and Nursing

What is the rule for sterile technique when placing a urinary catheter?

Answer

Use sterile gloves, a sterile catheter and lubricant, clean the genital area first, and never force the catheter.

#25 · Animal Care and Nursing

Where should the collection bag of an indwelling urinary catheter be kept?

Answer

Below the level of the bladder so urine drains by gravity, with a closed collection system.

#26 · Animal Care and Nursing

What is normal urine output in a hydrated dog or cat?

Answer

About 1 to 2 mL per kg per hour. Under 1 mL per kg per hour on fluids is a warning sign.

#27 · Animal Care and Nursing

What are the 4 phases of wound healing?

Answer

Inflammation, debridement, repair (new tissue) and maturation.

#28 · Animal Care and Nursing

What are the 4 wound classes?

Answer

Clean, clean-contaminated, contaminated and dirty.

#29 · Animal Care and Nursing

What is the golden period of a wound?

Answer

About 6 hours, after which bacteria multiply enough that the wound is treated as contaminated.

#30 · Animal Care and Nursing

What do you do first for a wound before clipping the hair?

Answer

Fill the wound with sterile water-soluble gel or cover it with a sterile moist dressing so hair stays out.

#31 · Animal Care and Nursing

What fluid is used for wound lavage?

Answer

Warm sterile saline or lactated Ringer's solution. Chlorhexidine diluted to 0.05 percent is also safe.

#32 · Animal Care and Nursing

How is a safe lavage pressure of about 7 to 8 psi made?

Answer

With a 35 mL syringe and an 18 gauge needle.

#33 · Animal Care and Nursing

Which wound dressing is used in the debridement phase?

Answer

An adherent wet-to-dry dressing, which pulls off dead tissue when removed.

#34 · Animal Care and Nursing

Which wound dressing is used in the repair phase?

Answer

A nonadherent moist dressing so new tissue is not torn off.

#35 · Animal Care and Nursing

What are the 3 layers of a bandage?

Answer

Primary (contact) layer, secondary (padding) layer and tertiary (outer holding) layer.

#36 · Animal Care and Nursing

Why do you leave the toes of a bandaged limb visible?

Answer

So you can check for swelling, coolness or colour change that shows the bandage is too tight.

#37 · Animal Care and Nursing

In which direction is a limb bandage applied?

Answer

From the distal end (toes) up to the proximal end so swelling is not trapped below the wrap.

#38 · Animal Care and Nursing

What signs suggest a bandage is too tight?

Answer

Swollen or spread toes, a cold limb, and a patient that licks or chews the bandage.

#39 · Animal Care and Nursing

When must you change a bandage?

Answer

When it is wet, soiled, slipping or smelly, and at least as often as the veterinarian orders.

#40 · Animal Care and Nursing

What is the maintenance fluid rate for dogs and cats?

Answer

About 40 to 60 mL per kg per day.

#41 · Animal Care and Nursing

How do you calculate a fluid deficit in mL?

Answer

Percent dehydration as a decimal x body weight in kg x 1000. For 8 percent in 10 kg that is 800 mL.

#42 · Animal Care and Nursing

How do you calculate drops per minute for an IV drip?

Answer

(mL per hour x drops per mL of the set) divided by 60.

#43 · Animal Care and Nursing

A 20 kg dog needs 60 mL per kg per day. What is the hourly rate?

Answer

20 x 60 = 1200 mL per day, and 1200 divided by 24 is 50 mL per hour.

#44 · Animal Care and Nursing

With a 15 drops per mL set, what drip rate gives 50 mL per hour?

Answer

(50 x 15) divided by 60 = 12.5 drops per minute.

#45 · Animal Care and Nursing

How many drops per mL does a pediatric (micro) drip set deliver?

Answer

60 drops per mL.

#46 · Animal Care and Nursing

What signs suggest fluid overload?

Answer

Serous nasal discharge, chemosis, coughing, a rising respiratory rate and weight gain.

#47 · Animal Care and Nursing

How often should a recumbent patient be turned?

Answer

About every 4 hours, with padding on bony points and clean, dry bedding.

#48 · Animal Care and Nursing

When are cold packs used on an injury?

Answer

In the first 24 to 72 hours to reduce swelling. Heat is used later to relax muscle.

#49 · Animal Care and Nursing

Why does a cuff that is too narrow give a false blood pressure reading?

Answer

It reads falsely high. A cuff that is too wide reads falsely low.

#50 · Animal Care and Nursing

What cuff width is correct for blood pressure measurement?

Answer

About 30 to 40 percent of the limb circumference.

#51 · Animal Care and Nursing

What does the Doppler method measure in a dog or cat?

Answer

Systolic blood pressure, using an ultrasound probe over an artery with a cuff.

#52 · Animal Care and Nursing

What systolic pressure in a dog or cat is considered hypertension?

Answer

Above about 160 mmHg.

#53 · Animal Care and Nursing

What mean arterial pressure shows organs may not be perfused?

Answer

A mean below about 60 mmHg.

#54 · Animal Care and Nursing

How should the cuff be positioned during blood pressure measurement?

Answer

At heart level, on a calm patient, with several readings averaged.

#55 · Animal Care and Nursing

In what position is a patient placed for an ECG?

Answer

Right lateral recumbency with the limbs parallel and relaxed.

#56 · Animal Care and Nursing

What are the standard ECG lead colours?

Answer

White right front, black left front, red left rear and green right rear.

#57 · Animal Care and Nursing

Which ECG lead is used to evaluate rhythm?

Answer

Lead II.

#58 · Animal Care and Nursing

What are standard ECG paper speeds and calibration?

Answer

25 or 50 mm per second, with a calibration of 10 mm equal to 1 mV.

#59 · Animal Care and Nursing

How do you find heart rate from a 6 second ECG strip?

Answer

Count the complexes and multiply by 10.

#60 · Animal Care and Nursing

Which patient actions cause ECG artifact?

Answer

Purring, panting, shivering and movement.

#61 · Animal Care and Nursing

What does tonometry measure?

Answer

Intraocular pressure. Normal in dogs and cats is about 10 to 25 mmHg.

#62 · Animal Care and Nursing

What condition does high intraocular pressure suggest?

Answer

Glaucoma, a painful emergency.

#63 · Animal Care and Nursing

Why must you avoid pressure on the globe during tonometry?

Answer

Pressing the eyelids against the globe falsely raises the reading.

#64 · Animal Care and Nursing

What does the Schirmer tear test measure and what is normal in a dog?

Answer

Tear production, normally 15 mm per minute or more in dogs.

#65 · Animal Care and Nursing

When is the Schirmer test done relative to eye drops?

Answer

Before any drops or fluorescein are put in the eye.

#66 · Animal Care and Nursing

What does fluorescein stain show?

Answer

A corneal ulcer, which stains green. Rinse the eye afterward.

#67 · Animal Care and Nursing

What is a normal SpO2 on room air?

Answer

95 percent or higher.

#68 · Animal Care and Nursing

What are the rights of medication administration?

Answer

Right patient, drug, dose, route, time and documentation.

#69 · Animal Care and Nursing

Which sites are used for IM injection in a dog or cat?

Answer

Quadriceps, lumbar epaxial muscles and hamstring muscles, avoiding the sciatic nerve.

#70 · Animal Care and Nursing

Why do you pull back on the plunger before an IM injection?

Answer

To check that the needle is not in a blood vessel.

#71 · Animal Care and Nursing

Which route gives the fastest drug effect?

Answer

Intravenous.

#72 · Animal Care and Nursing

What is extravasation?

Answer

Leakage of an IV drug out of the vein into surrounding tissue, which can cause tissue damage.

#73 · Animal Care and Nursing

Which veins are used for IV catheters in dogs and cats?

Answer

Cephalic, lateral saphenous and jugular.

#74 · Animal Care and Nursing

What is the order of eye medications when giving both drops and ointment?

Answer

Drops first, then ointment after about 5 minutes.

#75 · Animal Care and Nursing

Why give a water flush after a pill in a cat?

Answer

To help it pass into the stomach and prevent esophageal stricture, for example from doxycycline.

#76 · Animal Care and Nursing

Which are the recommended feline vaccine sites?

Answer

FVRCP right front limb, rabies right rear limb and FeLV left rear limb.

#77 · Animal Care and Nursing

How do you calculate the volume of a drug to give?

Answer

Volume in mL = dose in mg per kg x weight in kg divided by concentration in mg per mL.

#78 · Animal Care and Nursing

A 24 kg dog needs 2 mg per kg and the drug is 50 mg per mL. What volume?

Answer

2 x 24 = 48 mg, and 48 divided by 50 = 0.96 mL.

#79 · Animal Care and Nursing

How do you convert pounds to kilograms?

Answer

Divide by 2.2. For 22 lb that is 10 kg.

#80 · Animal Care and Nursing

How many mg per mL is a 1 percent solution?

Answer

10 mg per mL, since 1 percent is 1 g per 100 mL.

#81 · Animal Care and Nursing

How do you dilute 50 percent dextrose to 5 percent?

Answer

Mix 1 part of the 50 percent dextrose with 9 parts of fluid, a 1 to 10 dilution.

#82 · Animal Care and Nursing

Which Canadian law controls opioids, ketamine and benzodiazepines?

Answer

The Controlled Drugs and Substances Act. Keep these locked with a log of every dose and a regular count.

#83 · Animal Care and Nursing

What do you do if a controlled drug count is off?

Answer

Report the discrepancy to the veterinarian right away.

#84 · Animal Care and Nursing

What is the blood tube additive for a CBC?

Answer

EDTA, in a lavender top tube.

#85 · Animal Care and Nursing

Which tube is used for coagulation tests and what is its fill rule?

Answer

Light blue top (sodium citrate), filled exactly to the line for a 9 to 1 blood to citrate ratio.

#86 · Animal Care and Nursing

Which tube preserves glucose?

Answer

Grey top, which contains fluoride.

#87 · Animal Care and Nursing

When drawing several tubes from one venipuncture on a hospitalized patient, which tube is filled first?

Answer

A blood culture bottle comes first when collected, so later additives cannot contaminate it; then citrate, serum, heparin, EDTA and fluoride.

#88 · Animal Care and Nursing

How much blood can a healthy animal safely lose in one collection?

Answer

About 10 mL per kg of body weight.

#89 · Animal Care and Nursing

How is capillary refill time (CRT) checked and what is normal in a dog or cat?

Answer

Press the gum until it blanches, release, and time the return of pink colour. Normal CRT is under 2 seconds; prolonged CRT suggests poor perfusion and a slow return.

#90 · Animal Care and Nursing

What is the difference between cleaning and disinfecting?

Answer

Cleaning removes dirt and organic matter. Disinfecting kills microbes and works only on a clean surface.

#91 · Animal Care and Nursing

What disinfectant dilution is commonly used for canine parvovirus?

Answer

Bleach at about 1 part in 32, with the label contact time. Parvovirus survives for months.

#92 · Animal Care and Nursing

Why must bleach never be mixed with ammonia or acids?

Answer

Toxic gas forms.

#93 · Animal Care and Nursing

In what order do you care for patients in an infectious disease outbreak?

Answer

Healthy patients first and infectious patients last, with separate gloves and gowns.

#94 · Animal Care and Nursing

What is the rule on recapping needles?

Answer

Do not recap a needle with two hands. Dispose of sharps in a puncture proof container.

#95 · Animal Care and Nursing

Who handles reportable and immediately notifiable animal diseases in Canada?

Answer

The Canadian Food Inspection Agency. Tell the veterinarian right away if you suspect one.

#96 · Animal Care and Nursing

Which body signals show a fearful dog?

Answer

Ears back, tucked tail, lip licking, yawning, whale eye and turning away.

#97 · Animal Care and Nursing

When should a muzzle not be used on a dog?

Answer

When the dog is vomiting, is having trouble breathing, or is brachycephalic without close monitoring.

#98 · Animal Care and Nursing

What is the safe way to restrain a rabbit?

Answer

Support the hindquarters at all times and wrap in a towel, since a kick can fracture the spine.

#99 · Animal Care and Nursing

Why must you never compress the keel of a bird?

Answer

Birds have no diaphragm and breathe by moving the chest, so compression stops breathing.

#100 · Animal Care and Nursing

Where do you stand when approaching a horse and why?

Answer

At the left shoulder with a hand on the animal, not behind it, to avoid the kick zone.

#101 · Pharmacy and Pharmacology

What is bioavailability?

Answer

The fraction of a dose that reaches the systemic circulation. IV gives 100 percent because it skips absorption.

#102 · Pharmacy and Pharmacology

Which route gives the fastest onset?

Answer

Intravenous. It skips absorption, so the drug is in the circulation at once.

#103 · Pharmacy and Pharmacology

Why is IV preferred over SQ or IM in a patient in shock?

Answer

Poor perfusion slows absorption from muscle and skin, so IM and SQ drugs act slowly and unpredictably.

#104 · Pharmacy and Pharmacology

What is the first-pass effect?

Answer

An oral drug travels by the portal vein to the liver first, where part of it may be broken down before reaching the rest of the body.

#105 · Pharmacy and Pharmacology

Why can low albumin make a normal dose act like an overdose?

Answer

Many drugs bind albumin and only the unbound part is active, so less binding leaves more active drug.

#106 · Pharmacy and Pharmacology

Which species lacks glucuronidation, and what drug is the classic danger?

Answer

Cats. Acetaminophen is toxic to cats, and aspirin clears very slowly in them.

#107 · Pharmacy and Pharmacology

Which gene mutation makes collies sensitive to ivermectin and loperamide?

Answer

The MDR1 (ABCB1) mutation, which lets drugs enter the brain and cause neurologic signs.

#108 · Pharmacy and Pharmacology

Why must a dog flea product containing permethrin never be used on a cat?

Answer

Pyrethroids can cause tremors and seizures in cats.

#109 · Pharmacy and Pharmacology

Why must doxycycline tablets be followed by water in cats?

Answer

The tablet can lodge in the esophagus and cause esophagitis and strictures.

#110 · Pharmacy and Pharmacology

When is the intraosseous route used?

Answer

When IV access cannot be obtained in a small or collapsed patient. The veterinarian or a supervised person places it.

#111 · Pharmacy and Pharmacology

What are the five rights of medication?

Answer

Right patient, drug, dose, route and time. Right documentation is added as a sixth.

#112 · Pharmacy and Pharmacology

At what points should you check a drug label?

Answer

When taking it from storage, when drawing it up, and before giving it.

#113 · Pharmacy and Pharmacology

What does a complete drug order contain?

Answer

The patient, drug, dose, route, frequency and duration. If any part is unclear, ask the veterinarian before acting.

#114 · Pharmacy and Pharmacology

What do OD, OS and OU mean?

Answer

OD is the right eye, OS the left eye, and OU both eyes.

#115 · Pharmacy and Pharmacology

What is the usual concentration of veterinary insulin and the matching syringe?

Answer

U-40 (40 units per mL) with a U-40 syringe. Human U-100 insulin needs a U-100 syringe.

#116 · Pharmacy and Pharmacology

What happens if a U-100 syringe is used for U-40 insulin?

Answer

The animal receives well under half the intended dose, so the syringe must always match the insulin.

#117 · Pharmacy and Pharmacology

How many mg/mL is a 1 percent solution?

Answer

10 mg/mL, because 1 g is in 100 mL. Multiply the percent by 10.

#118 · Pharmacy and Pharmacology

How many mg/mL is a 2 percent lidocaine solution?

Answer

20 mg/mL.

#119 · Pharmacy and Pharmacology

What concentration is a 1:1,000 solution?

Answer

1 mg/mL, because 1 g is in 1,000 mL. A 1:10,000 solution is 0.1 mg/mL.

#120 · Pharmacy and Pharmacology

Give the four steps of a dose calculation.

Answer

Convert weight to kg, multiply mg/kg by kg for the dose in mg, divide by mg/mL for the volume, then check that the answer is reasonable.

#121 · Pharmacy and Pharmacology

A 10 kg dog needs 1 mg/kg of a drug at 10 mg/mL. What volume do you draw up?

Answer

Dose is 10 mg, and 10 mg divided by 10 mg/mL = 1 mL.

#122 · Pharmacy and Pharmacology

A 4.5 kg cat needs 0.02 mg/kg buprenorphine at 0.3 mg/mL. What volume?

Answer

0.02 x 4.5 = 0.09 mg, and 0.09 divided by 0.3 = 0.3 mL.

#123 · Pharmacy and Pharmacology

A calculated dose gives 1.25 tablets. What do you do?

Answer

Tell the veterinarian, who decides on another strength or a rounded dose. Do not round on your own.

#124 · Pharmacy and Pharmacology

What equation is used for dilutions?

Answer

C1 x V1 = C2 x V2.

#125 · Pharmacy and Pharmacology

How much 50 percent dextrose makes 100 mL of 5 percent dextrose?

Answer

50 x V1 = 5 x 100, so V1 = 10 mL, added to 90 mL of diluent.

#126 · Pharmacy and Pharmacology

Why is 5 percent dextrose in water not used for resuscitation?

Answer

The dextrose is metabolized, leaving free water, so it does not expand the vascular volume.

#127 · Pharmacy and Pharmacology

What is the usual teaching range for maintenance fluids in dogs and cats?

Answer

About 40 to 60 mL/kg/day, with 60 mL/kg/day a common figure. The veterinarian's order rules.

#128 · Pharmacy and Pharmacology

What is the dehydration deficit formula?

Answer

Body weight in kg x percent dehydration as a decimal x 1,000 = mL.

#129 · Pharmacy and Pharmacology

A 10 kg dog is 8 percent dehydrated. What is the deficit?

Answer

10 x 0.08 x 1,000 = 800 mL.

#130 · Pharmacy and Pharmacology

What is the maintenance rate in mL/hr for a 10 kg dog at 60 mL/kg/day?

Answer

600 mL per day divided by 24 = 25 mL/hr.

#131 · Pharmacy and Pharmacology

What is the typical shock bolus range for dogs and cats?

Answer

Up to 90 mL/kg in dogs and 40 to 60 mL/kg in cats, given in aliquots of one quarter to one third, then reassess.

#132 · Pharmacy and Pharmacology

List signs of fluid overload.

Answer

Serous nasal discharge, chemosis, increased respiratory rate and effort, cough and peripheral edema. Slow or stop fluids and tell the veterinarian.

#133 · Pharmacy and Pharmacology

What is the drip rate formula?

Answer

Drops per minute = volume in mL x drip set in drops/mL divided by time in minutes.

#134 · Pharmacy and Pharmacology

Deliver 500 mL over 4 hours with a 15 drops/mL set. What is the rate?

Answer

500 x 15 = 7,500 divided by 240 minutes = about 31 drops per minute.

#135 · Pharmacy and Pharmacology

What are the steps of a CRI calculation in mcg/kg/min?

Answer

Rate x kg gives mcg/min, multiply by 60 for mcg/hr, then divide by the concentration in mcg/mL to get mL/hr.

#136 · Pharmacy and Pharmacology

A 20 kg dog gets lidocaine 50 mcg/kg/min from a 2 percent solution. What is the rate?

Answer

1,000 mcg/min x 60 = 60,000 mcg/hr, divided by 20,000 mcg/mL = 3 mL/hr.

#137 · Pharmacy and Pharmacology

What is the maximum rate of potassium delivery in fluids?

Answer

0.5 mEq/kg/hr. Never give concentrated potassium as an IV push.

#138 · Pharmacy and Pharmacology

What are the main risks of NSAIDs such as carprofen and meloxicam?

Answer

Gastrointestinal ulceration and kidney injury, and less often liver injury. Do not combine with corticosteroids or another NSAID.

#139 · Pharmacy and Pharmacology

Which common human drug is toxic to cats and should never be given?

Answer

Acetaminophen, which damages red blood cells and causes methemoglobinemia.

#140 · Pharmacy and Pharmacology

Which opioid is a partial agonist and which is an agonist-antagonist?

Answer

Buprenorphine is a partial agonist and butorphanol is an agonist-antagonist.

#141 · Pharmacy and Pharmacology

What are the common side effects of opioids?

Answer

Sedation, respiratory depression, bradycardia, panting, vomiting and constipation. Cats may show excitement at high doses.

#142 · Pharmacy and Pharmacology

What reverses opioids, and what is the drawback?

Answer

Naloxone. It also reverses the pain relief.

#143 · Pharmacy and Pharmacology

What are the cardiovascular effects of alpha-2 agonists?

Answer

Bradycardia, vasoconstriction and decreased cardiac output.

#144 · Pharmacy and Pharmacology

Which drugs reverse dexmedetomidine and xylazine?

Answer

Atipamezole reverses dexmedetomidine, and yohimbine can reverse xylazine.

#145 · Pharmacy and Pharmacology

What is the main drawback of acepromazine?

Answer

It causes hypotension and gives no analgesia, so use care in hypovolemic patients.

#146 · Pharmacy and Pharmacology

What are expected side effects of prednisone?

Answer

Increased thirst, urination and appetite, with infection risk and skin thinning in long courses. Do not stop long-term steroids suddenly.

#147 · Pharmacy and Pharmacology

Why is prednisolone usually used in cats instead of prednisone?

Answer

Cats convert prednisone poorly to its active form.

#148 · Pharmacy and Pharmacology

What are the risks of fluoroquinolones?

Answer

Cartilage damage in growing dogs and blindness in cats at high doses.

#149 · Pharmacy and Pharmacology

What has been required in Canada for medically important antimicrobials since December 2018?

Answer

A veterinary prescription.

#150 · Pharmacy and Pharmacology

What is the antidote for anticoagulant rodenticide?

Answer

Vitamin K1.

#151 · Pharmacy and Pharmacology

What is the antidote for ethylene glycol in dogs?

Answer

Fomepizole, given early. Antifreeze causes kidney failure.

#152 · Pharmacy and Pharmacology

What does xylitol cause in dogs?

Answer

Severe low blood sugar and liver injury.

#153 · Pharmacy and Pharmacology

What epinephrine dose is used in anaphylaxis, and what volume is that per 10 kg of a 1 mg/mL solution?

Answer

0.01 mg/kg, which is 0.1 mL per 10 kg. Ordered by the veterinarian.

#154 · Pharmacy and Pharmacology

What must appear on a dispensed medication label?

Answer

Clinic, client and patient, drug name, strength and quantity, directions, date, expiry, veterinarian and any cautions.

#155 · Pharmacy and Pharmacology

What is a withdrawal time?

Answer

The period after the last dose before meat, milk or eggs from a treated food animal may be sold.

#156 · Pharmacy and Pharmacology

Which Canadian law governs opioids, ketamine, benzodiazepines and barbiturates?

Answer

The federal Controlled Drugs and Substances Act (CDSA) and its regulations.

#157 · Pharmacy and Pharmacology

What does a controlled drug log entry record?

Answer

Date, patient and client, drug, amount used and wasted, who gave it, who ordered it and the new running balance.

#158 · Pharmacy and Pharmacology

How do you correct an error in a controlled drug logbook?

Answer

Draw a single line through it and initial it. Never erase or cover it.

#159 · Pharmacy and Pharmacology

What do you do if a controlled drug count does not match the logbook?

Answer

Report at once to the supervising veterinarian so it is investigated. Loss or theft is reported to Health Canada and police as required.

#160 · Pharmacy and Pharmacology

What does WHMIS 2015 use to communicate chemical hazards?

Answer

Hazard pictograms, supplier labels and Safety Data Sheets, which have 16 sections.

#161 · Pharmacy and Pharmacology

At what temperature are refrigerated drugs and vaccines kept, and what must never happen?

Answer

2 to 8 degrees C, and they must not be frozen.

#162 · Pharmacy and Pharmacology

Where should vaccines be given in cats, and why?

Answer

In the distal limb, with the site recorded, because of injection-site sarcoma risk.

#163 · Pharmacy and Pharmacology

What PPE and waste routes apply to chemotherapy drugs?

Answer

Chemotherapy-rated gloves, gown and eye protection, with waste in a labelled cytotoxic container. Pregnant staff should avoid handling.

#164 · Pharmacy and Pharmacology

How should unwanted drugs be disposed of?

Answer

By the clinic's pharmaceutical waste or take-back arrangement. Never flush them.

#165 · Pharmacy and Pharmacology

What do you do first when an order is unclear or looks unsafe?

Answer

Ask the veterinarian before acting.

#166 · Surgical Nursing

Which layer of the ventral abdominal wall is the main holding layer in a midline closure?

Answer

The linea alba, the avascular fibrous midline. It is closed with suture that keeps its strength until healing is secure.

#167 · Surgical Nursing

What is removed in an ovariohysterectomy?

Answer

Both ovaries and the uterus. The ovarian pedicles are ligated and the uterine body is ligated just cranial to the cervix.

#168 · Surgical Nursing

What structures in the ovarian pedicle must be ligated to prevent hemorrhage?

Answer

The ovarian artery and vein, held within the ovarian pedicle.

#169 · Surgical Nursing

What does the spermatic cord contain that must be ligated in a castration?

Answer

The testicular vessels and the ductus deferens.

#170 · Surgical Nursing

Name four patient factors that slow wound healing.

Answer

Poor nutrition or low protein, anemia, steroid use or diabetes, obesity, infection and hypothermia are all taught as slowing healing.

#171 · Surgical Nursing

Why does hypothermia matter to a surgical patient?

Answer

A cold patient clots poorly, heals slowly, recovers slowly and has a higher infection risk.

#172 · Surgical Nursing

Classify a routine spay by contamination.

Answer

Clean: no inflammation and no entry into the gastrointestinal, respiratory or urinary tract.

#173 · Surgical Nursing

Classify a cystotomy with sterile urine.

Answer

Clean-contaminated: a hollow organ is entered under control with minimal spillage.

#174 · Surgical Nursing

Classify surgery on an existing abscess or ruptured bowel.

Answer

Dirty: infection already exists at the site.

#175 · Surgical Nursing

What are the phases of wound healing in order?

Answer

Inflammatory (debridement), repair (proliferative), then maturation (remodelling).

#176 · Surgical Nursing

How long does the inflammatory phase of wound healing last?

Answer

About the first 3 to 5 days.

#177 · Surgical Nursing

When does granulation tissue begin forming?

Answer

About day 3 to 5, during the repair phase, when fibroblasts lay down collagen.

#178 · Surgical Nursing

What is healing by first intention?

Answer

Primary closure: the wound edges are brought together right away.

#179 · Surgical Nursing

What is delayed primary closure?

Answer

The wound is left open for a few days, often bandaged, and closed once infection risk has fallen.

#180 · Surgical Nursing

What is the golden period for a traumatic wound?

Answer

The first few hours, commonly taught as about 6, when the wound can usually still be closed.

#181 · Surgical Nursing

When does incision dehiscence most often occur?

Answer

Around days 3 to 5, especially in an overactive patient or after suture failure.

#182 · Surgical Nursing

What prevents most self-trauma to a surgical incision?

Answer

An Elizabethan collar plus restricted exercise.

#183 · Surgical Nursing

Which scalpel blades fit a No. 3 Bard-Parker handle?

Answer

Blades 10, 11, 12 and 15.

#184 · Surgical Nursing

Which scalpel handle takes blades 20 to 25?

Answer

The No. 4 handle.

#185 · Surgical Nursing

How should a scalpel blade be removed from its handle?

Answer

With a hemostat or needle holder, never by hand.

#186 · Surgical Nursing

Which scissors cut suture and heavy tissue, Mayo or Metzenbaum?

Answer

Mayo scissors. Metzenbaum scissors are light and for dissecting delicate tissue only.

#187 · Surgical Nursing

What is the common needle holder with a locking ratchet?

Answer

The Mayo-Hegar needle holder.

#188 · Surgical Nursing

Which hemostats are larger and used on pedicles?

Answer

Rochester-Carmalt forceps. Mosquito hemostats are the smallest.

#189 · Surgical Nursing

Which tissue forceps is atraumatic and used on intestine?

Answer

The Babcock forceps. Allis forceps have teeth and are traumatic.

#190 · Surgical Nursing

What do Backhaus towel clamps do?

Answer

Hold drapes in place by piercing them.

#191 · Surgical Nursing

Name two self-retaining retractors.

Answer

Balfour and Gelpi. Senn and Army-Navy retractors are hand-held.

#192 · Surgical Nursing

How should hinged instruments be wrapped for sterilization?

Answer

With hinges open and ratchets unlocked so steam reaches all surfaces.

#193 · Surgical Nursing

Why rinse instruments right after use?

Answer

Dried blood and protein shield microbes from sterilization, and hot water bakes protein on.

#194 · Surgical Nursing

How does an ultrasonic cleaner work?

Answer

It uses cavitation, tiny bubbles, to dislodge debris. Do not mix metals or use plain saline.

#195 · Surgical Nursing

What should hinged instruments be lubricated with?

Answer

A water-soluble instrument lubricant ('milk'), not oil.

#196 · Surgical Nursing

Why avoid saline and bleach soaks on stainless steel instruments?

Answer

Chlorides cause pitting and corrosion.

#197 · Surgical Nursing

What is the difference between sterilization and disinfection?

Answer

Sterilization kills all microorganisms including spores. Disinfection does not.

#198 · Surgical Nursing

What are the standard conditions for steam sterilizing a wrapped pack?

Answer

121 degrees C (250 degrees F) at about 15 psi for 30 minutes. Follow the manufacturer for other cycles.

#199 · Surgical Nursing

Which items cannot go in a steam autoclave?

Answer

Heat-sensitive or moisture-sensitive items such as many plastics and electronics.

#200 · Surgical Nursing

Name two hazards of ethylene oxide gas.

Answer

It is toxic and a carcinogen, so items must be aerated and the area ventilated.

#201 · Surgical Nursing

What type of items suit ethylene oxide or gas plasma sterilization?

Answer

Heat-sensitive items such as plastics and endoscopes.

#202 · Surgical Nursing

What does external process indicator tape prove?

Answer

Only that the pack went through a cycle, not that it is sterile.

#203 · Surgical Nursing

Where is an internal chemical indicator placed?

Answer

In the centre of the pack, to show steam reached the middle.

#204 · Surgical Nursing

Which test actually proves a cycle sterilized the load?

Answer

A biological indicator containing spores.

#205 · Surgical Nursing

Which spore organism is used for steam biological indicators?

Answer

Geobacillus stearothermophilus. Bacillus atrophaeus is used for ethylene oxide and dry heat.

#206 · Surgical Nursing

What do you do if a biological indicator is positive?

Answer

Take the load out of use, recall what you can and troubleshoot before the next cycle.

#207 · Surgical Nursing

What does event-related shelf life mean?

Answer

A pack stays sterile until an event compromises it, such as a tear, moisture or a drop.

#208 · Surgical Nursing

What should you do with a sterilized pack that is wet or torn?

Answer

Treat it as unsterile and resterilize it.

#209 · Surgical Nursing

About how long are healthy adult dogs and cats usually fasted before anesthesia?

Answer

Roughly 4 to 6 hours, with water until premedication, as the veterinarian directs.

#210 · Surgical Nursing

Why are rabbits and rodents not fasted before surgery?

Answer

They cannot vomit.

#211 · Surgical Nursing

What starting fluid rate during anesthesia is commonly taught for dogs and for cats?

Answer

About 5 mL/kg/hr for dogs and 3 mL/kg/hr for cats, adjusted by the veterinarian.

#212 · Surgical Nursing

A 44 lb dog needs fluids at 5 mL/kg/hr. What is the rate?

Answer

44 lb divided by 2.2 is 20 kg, and 20 times 5 is 100 mL/hr.

#213 · Surgical Nursing

Where is the surgical site clipped and why?

Answer

In the prep room, away from the surgical suite, so hair does not spread into the room.

#214 · Surgical Nursing

In what pattern is the surgical scrub applied?

Answer

From the centre outward in a spiral, never returning to the centre with the same sponge.

#215 · Surgical Nursing

How do you dilute 2 percent chlorhexidine to 0.05 percent?

Answer

The factor is 2 divided by 0.05, which is 40, so 1 part stock plus 39 parts water.

#216 · Surgical Nursing

How are the hands held during a surgical hand scrub?

Answer

Hands up and elbows down so water runs from the fingertips to the elbows.

#217 · Surgical Nursing

Which part of a sterile gown is considered sterile?

Answer

The front from chest to waist and the sleeves, except the cuffs and above the elbow. The back is not sterile.

#218 · Surgical Nursing

At what level is a sterile table considered sterile?

Answer

Only at table height. Anything below the table edge is not sterile.

#219 · Surgical Nursing

How much of a drape's edge is non-sterile?

Answer

The outer 2.5 cm (about 1 inch).

#220 · Surgical Nursing

What is strike-through?

Answer

Moisture wicking microbes through a wet barrier, making the barrier contaminated.

#221 · Surgical Nursing

What should a scrubbed person do if unsure whether an item is sterile?

Answer

Treat it as not sterile: when in doubt, throw it out.

#222 · Surgical Nursing

What happens to a gloved hand that drops below the waist?

Answer

It is considered contaminated, so change the glove.

#223 · Surgical Nursing

What does 4-0 suture mean compared with 3-0?

Answer

More zeros means a smaller diameter, so 4-0 is thinner than 3-0.

#224 · Surgical Nursing

Which suture type harbours bacteria more readily, monofilament or braided?

Answer

Braided (multifilament) suture, which can wick bacteria. Monofilament passes smoothly.

#225 · Surgical Nursing

Which needle suits soft tissue such as intestine and bladder?

Answer

A taper (round body) needle, which pierces without cutting.

#226 · Surgical Nursing

Which suture pattern folds tissue into a hollow organ?

Answer

An inverting pattern such as Lembert, Cushing or Connell.

#227 · Surgical Nursing

What suture length is commonly taught for a continuous closure?

Answer

About 4 times the length of the incision.

#228 · Surgical Nursing

What are the circulating technician's counting duties?

Answer

Count sponges, needles and blades with the scrub person before the incision and before closure, and report any mismatch before closure.

#229 · Surgical Nursing

How must controlled drugs used with anesthesia be handled in Canada?

Answer

They fall under the Controlled Drugs and Substances Act, so keep the logbook and counts accurate.

#230 · Surgical Nursing

What is a seroma?

Answer

A soft pocket of serum under the skin, usually not painful.

#231 · Anesthesia

What four things does general anesthesia provide?

Answer

Unconsciousness, muscle relaxation, loss of pain perception and loss of protective reflexes.

#232 · Anesthesia

What is balanced anesthesia?

Answer

Using several drugs at lower doses so each drug's side effects stay smaller.

#233 · Anesthesia

What is the formula for blood pressure?

Answer

Blood pressure equals cardiac output times systemic vascular resistance.

#234 · Anesthesia

What MAP is the minimum target under anesthesia?

Answer

At least 60 mmHg, to keep the kidneys and brain perfused. Normal MAP is about 80 to 100 mmHg.

#235 · Anesthesia

What is the normal EtCO2 range?

Answer

35 to 45 mmHg. Above about 50 to 60 mmHg means hypoventilation.

#236 · Anesthesia

At what SpO2 do you act immediately?

Answer

Below 90 percent is hypoxemia. The target on oxygen is 95 percent or higher.

#237 · Anesthesia

Why do anesthetized patients become hypothermic?

Answer

Anesthesia stops shivering and warming behaviour, and small patients lose heat fastest. A heating pad must never touch skin directly.

#238 · Anesthesia

What is ASA I?

Answer

A normal healthy patient.

#239 · Anesthesia

What is ASA II?

Answer

Mild systemic disease, for example mild obesity or a compensated heart murmur.

#240 · Anesthesia

What is ASA III?

Answer

Severe systemic disease, for example uncontrolled diabetes.

#241 · Anesthesia

What is ASA IV?

Answer

Severe disease that is a constant threat to life, for example shock.

#242 · Anesthesia

What does the letter E added to an ASA grade mean?

Answer

The case is an emergency, for example ASA IIIE.

#243 · Anesthesia

Which patients are not fasted for long before anesthesia?

Answer

Neonates, pediatric patients, very small patients, rabbits and rodents, because of the risk of low blood sugar. Rabbits and rodents also cannot vomit.

#244 · Anesthesia

Why is an IV catheter placed in nearly every anesthetized patient?

Answer

It gives fast access for emergency drugs and fluids.

#245 · Anesthesia

What does an anticholinergic do during anesthesia?

Answer

It blocks vagal tone, raising heart rate and drying secretions. Atropine and glycopyrrolate are examples.

#246 · Anesthesia

Does acepromazine provide pain relief?

Answer

No. It tranquilizes but gives no analgesia, and it causes vasodilation and low blood pressure.

#247 · Anesthesia

Which patients need caution with acepromazine?

Answer

Dehydrated or shocked patients, boxers, giant breeds and ABCB1 (MDR1) mutation breeds such as collies.

#248 · Anesthesia

Name common side effects of opioids.

Answer

Respiratory depression, slow heart rate, panting and vomiting.

#249 · Anesthesia

What reverses an opioid?

Answer

Naloxone.

#250 · Anesthesia

What reverses dexmedetomidine or xylazine?

Answer

Atipamezole.

#251 · Anesthesia

What are the main cardiovascular effects of alpha-2 agonists?

Answer

A large drop in heart rate and vasoconstriction followed by lower cardiac output. Use mainly in healthy patients.

#252 · Anesthesia

What reverses a benzodiazepine?

Answer

Flumazenil.

#253 · Anesthesia

How is propofol given for induction?

Answer

Slowly IV to effect. Too fast a dose causes apnea.

#254 · Anesthesia

How long may an opened propofol syringe be used?

Answer

Only for the time on the label, usually within 6 hours, because it supports bacterial growth.

#255 · Anesthesia

Which induction drug is a dissociative anesthetic?

Answer

Ketamine, often combined with a benzodiazepine.

#256 · Anesthesia

What are the side effects of ketamine?

Answer

Rigid muscles, seizures if used alone and raised eye pressure. It keeps blood pressure up.

#257 · Anesthesia

Why avoid ketamine in a cat with urinary obstruction?

Answer

It is cleared mostly by the kidneys in cats, so effects are prolonged.

#258 · Anesthesia

Which induction agent suits cardiac or critically ill patients?

Answer

Etomidate. It can suppress adrenal function and cause myoclonus.

#259 · Anesthesia

What does MAC measure?

Answer

The minimum alveolar concentration that stops movement in 50 percent of patients when stimulated. It is a measure of potency.

#260 · Anesthesia

Which is more potent, isoflurane or sevoflurane?

Answer

Isoflurane. Its MAC is about 1.3 percent in dogs against about 2.4 percent for sevoflurane.

#261 · Anesthesia

Why does sevoflurane give faster changes in depth?

Answer

It has lower blood-gas solubility than isoflurane.

#262 · Anesthesia

Can a vaporizer be used with any inhalant?

Answer

No. Each vaporizer is built for one specific agent.

#263 · Anesthesia

What are the two steps of any drug calculation?

Answer

Dose in mg equals mg/kg times kg, then volume in mL equals mg divided by mg/mL.

#264 · Anesthesia

A dog weighs 25 kg and the order is butorphanol 0.2 mg/kg from a 10 mg/mL vial. What volume?

Answer

0.2 x 25 = 5 mg, then 5 / 10 = 0.5 mL.

#265 · Anesthesia

A cat weighs 11 lb. What is its weight in kg?

Answer

11 / 2.2 = 5 kg.

#266 · Anesthesia

A 5 kg cat is ordered ketamine 5 mg/kg and the vial is 100 mg/mL. What volume?

Answer

5 x 5 = 25 mg, then 25 / 100 = 0.25 mL.

#267 · Anesthesia

A 20 kg dog needs propofol 4 mg/kg from a 10 mg/mL vial. What is the full calculated volume?

Answer

4 x 20 = 80 mg, which is 8 mL. Give a fraction first and more only to effect.

#268 · Anesthesia

A 10 kg dog needs fentanyl 5 mcg/kg/hr from a bag of 50 mcg/mL. What rate?

Answer

5 x 10 = 50 mcg/hr, then 50 / 50 = 1 mL/hr.

#269 · Anesthesia

What are typical maintenance anesthetic fluid rates?

Answer

About 5 mL/kg/hr in dogs and 3 mL/kg/hr in cats. The veterinarian sets the final rate.

#270 · Anesthesia

A 20 kg dog is on 5 mL/kg/hr fluids. What is the rate?

Answer

5 x 20 = 100 mL/hr.

#271 · Anesthesia

What is the most common cause of a calculation error in anesthesia?

Answer

A tenfold decimal error or a wrong concentration. Always check the label and use kg.

#272 · Anesthesia

What does the oxygen flush valve do?

Answer

It sends high-flow oxygen past the vaporizer, which dilutes the anesthetic. Do not use it routinely.

#273 · Anesthesia

What does the APL (pop-off) valve do?

Answer

It releases excess gas from the circuit. Never leave it closed on a connected patient.

#274 · Anesthesia

How do you give a manual breath safely?

Answer

Close the APL valve, squeeze the bag to about 15 to 20 cm H2O, then reopen the valve.

#275 · Anesthesia

What is a circle system?

Answer

A rebreathing circuit where exhaled gas passes through a CO2 absorber and is breathed again. Used over about 3 to 5 kg.

#276 · Anesthesia

Which circuits suit patients under about 3 to 5 kg?

Answer

Non-rebreathing systems such as the Bain, Mapleson D and Jackson-Rees, which have no absorber.

#277 · Anesthesia

What is the minimum fresh gas flow for a Bain or Jackson-Rees circuit?

Answer

About 200 mL/kg/min.

#278 · Anesthesia

How do you know soda lime is exhausted?

Answer

It changes colour and inspired CO_2 rises.

#279 · Anesthesia

How do you perform a circuit leak test?

Answer

Close the APL valve, occlude the Y-piece, fill to 20 cm H2O and check the pressure holds for at least 10 seconds.

#280 · Anesthesia

How do you know a charcoal scavenger canister is full?

Answer

It has gained the weight on its label, usually 50 grams. It does not absorb nitrous oxide.

#281 · Anesthesia

Where should the tip of the ET tube sit?

Answer

About the level of the thoracic inlet.

#282 · Anesthesia

What happens if the ET tube is pushed in too far?

Answer

It enters one bronchus, usually the right, giving one-lung ventilation and low SpO2. Pull it back slightly.

#283 · Anesthesia

What is the best confirmation of ET tube placement?

Answer

A capnograph waveform with EtCO2 readings. Fog in the tube and chest rise also help.

#284 · Anesthesia

What does a missing EtCO2 and a bulging abdomen suggest after intubation?

Answer

The tube is in the esophagus. Remove it and try again.

#285 · Anesthesia

How is the ET cuff inflated?

Answer

Inflate while squeezing the bag until the leak stops at about 20 cm H2O. Overinflation can tear the trachea.

#286 · Anesthesia

What must be done to the cuff before moving an intubated patient?

Answer

Deflate it, because moving an inflated tube can injure the trachea.

#287 · Anesthesia

What is a special intubation risk in cats?

Answer

Laryngospasm. A drop of lidocaine on the larynx helps, and the tube is small.

#288 · Anesthesia

How should brachycephalic dogs be managed at recovery?

Answer

Keep them intubated until fully awake, with extra tubes and suction ready.

#289 · Anesthesia

How often are anesthetic values recorded?

Answer

At least every 5 minutes, and a person stays with the patient.

#290 · Anesthesia

What signs suggest a surgical plane in a dog?

Answer

Central slightly constricted pupil, slow palpebral reflex, relaxed jaw tone and no response to surgery.

#291 · Anesthesia

What does a blink and active jaw tone mean?

Answer

The patient is too light.

#292 · Anesthesia

What cuff width is used for blood pressure measurement?

Answer

About 40 percent of the limb circumference. Too narrow reads falsely high, too wide reads falsely low.

#293 · Anesthesia

What should you check first when the patient is too light?

Answer

Check the vaporizer and oxygen flow, then notify the veterinarian before adding drugs.

#294 · Anesthesia

What do you do first for apnea under anesthesia?

Answer

Stop the vaporizer, ventilate with 100 percent oxygen at 8 to 12 breaths per minute and tell the veterinarian.

#295 · Anesthesia

What do you do first for low SpO2?

Answer

Check the oxygen supply, tube position and circuit before anything else.

#296 · Laboratory Procedures

Which tube is used for a CBC, and why?

Answer

Lavender (EDTA). EDTA binds calcium to stop clotting and keeps cell shape. It is not suitable for chemistry or coagulation tests.

#297 · Laboratory Procedures

What is the blood to citrate ratio in a light blue tube?

Answer

9 parts blood to 1 part sodium citrate. An underfilled tube gives falsely prolonged clotting times.

#298 · Laboratory Procedures

Which tube type gives serum with no additive?

Answer

Red top. Let it clot about 30 minutes, then centrifuge and separate the serum.

#299 · Laboratory Procedures

Why does a gray top tube suit glucose samples?

Answer

Sodium fluoride blocks glycolysis, so the cells do not keep using glucose after collection.

#300 · Laboratory Procedures

What is the usual order of draw?

Answer

Blood culture, citrate, serum, heparin, EDTA, fluoride. This limits additive carry-over.

#301 · Laboratory Procedures

What happens to a serum sample contaminated with EDTA?

Answer

Potassium rises and calcium falls, because EDTA contains potassium and binds calcium.

#302 · Laboratory Procedures

What is the effect of hemolysis on a chemistry sample?

Answer

It can raise potassium, AST and phosphorus and interfere with colour-based tests. Forcing blood through a needle is a common cause.

#303 · Laboratory Procedures

How do you reduce lipemia in a blood sample?

Answer

Fast the patient 8 to 12 hours before collection.

#304 · Laboratory Procedures

What does yellow plasma in a spun hematocrit tube suggest?

Answer

Icterus, from high bilirubin. Pink plasma suggests hemolysis and cloudy plasma suggests lipemia.

#305 · Laboratory Procedures

What two identifiers belong on every tube and slide?

Answer

Patient name or ID and the owner's name, plus date and sample type, written at the bedside.

#306 · Laboratory Procedures

Within how long should serum or plasma be separated from the cells?

Answer

About 2 hours. Glucose falls if serum sits on cells because the cells keep using it.

#307 · Laboratory Procedures

What fixative and ratio are used for histopathology tissue?

Answer

10 percent neutral buffered formalin at about 10 parts to 1 part tissue, in pieces no thicker than 1 cm.

#308 · Laboratory Procedures

Why should blood smears not be shipped with formalin?

Answer

Formalin fumes ruin the Romanowsky stain. Ship smears air-dried and separate.

#309 · Laboratory Procedures

Which Canadian rules apply to couriering diagnostic specimens?

Answer

Transportation of Dangerous Goods requirements: leak-proof primary container, absorbent material and the outer packaging the reference lab specifies.

#310 · Laboratory Procedures

What is the normal PCV range for a dog and for a cat?

Answer

Dog 0.37 to 0.55 L/L (37 to 55 percent); cat 0.30 to 0.45 L/L (30 to 45 percent).

#311 · Laboratory Procedures

What does a low PCV with low total protein suggest?

Answer

Blood loss. A low PCV with normal or high protein suggests hemolysis or poor production.

#312 · Laboratory Procedures

What does a high PCV with high total protein suggest?

Answer

Dehydration, since both red cells and plasma protein are concentrated.

#313 · Laboratory Procedures

What is a good blood smear like?

Answer

A feathered edge and a monolayer where cells touch but do not overlap, air-dried fast and then stained.

#314 · Laboratory Procedures

What angle is the spreader slide held at when making a smear?

Answer

About 30 to 45 degrees, pushed in one smooth motion.

#315 · Laboratory Procedures

How do you estimate platelets on a smear?

Answer

Average platelets per oil field over 10 fields, then multiply by about 15 to 20 x10^9/L. Check for clumps.

#316 · Laboratory Procedures

What is the typical stress leukogram in a dog?

Answer

Mature neutrophilia, lymphopenia, eosinopenia and monocytosis.

#317 · Laboratory Procedures

What does a left shift mean?

Answer

Increased band neutrophils in blood, which is a sign of inflammation. Toxic change may be seen too.

#318 · Laboratory Procedures

How is a corrected reticulocyte percentage calculated?

Answer

Reticulocyte % x (patient PCV / normal PCV). Example: 6 x (20/45) = 2.67 percent.

#319 · Laboratory Procedures

What is the absolute reticulocyte count formula, and the dog regenerative cutoff?

Answer

Reticulocyte % x RBC count. A dog with an absolute count above about 60 x10^9/L is regenerative.

#320 · Laboratory Procedures

How do you correct a WBC count for nucleated red cells?

Answer

WBC x 100 / (100 + nRBC per 100 WBC). Example: 20 x 100 / 125 = 16 x10^9/L.

#321 · Laboratory Procedures

A dog has WBC 15 x10^9/L and 70 percent neutrophils. What is the absolute neutrophil count?

Answer

15 x 0.70 = 10.5 x10^9/L.

#322 · Laboratory Procedures

Which red cell finding points to immune-mediated hemolytic anemia in a dog?

Answer

Spherocytes and autoagglutination. A drop of blood with saline that stays clumped confirms agglutination.

#323 · Laboratory Procedures

What stain shows Heinz bodies, and in which species are they classic?

Answer

New methylene blue. They are classic in cats after oxidative injury such as acetaminophen.

#324 · Laboratory Procedures

Which coagulation value rises first with early rodenticide toxicity?

Answer

PT, because factor VII has the shortest half-life, so aPTT stays normal at first.

#325 · Laboratory Procedures

What do you convert glucose from mg/dL to mmol/L with?

Answer

Divide mg/dL by 18. A dog value of 90 mg/dL is 5.0 mmol/L.

#326 · Laboratory Procedures

How do you convert creatinine from mg/dL to umol/L?

Answer

Multiply by 88.4. A value of 1.5 mg/dL is about 133 umol/L.

#327 · Laboratory Procedures

Which instrument measures urine specific gravity reliably?

Answer

A refractometer, calibrated with distilled water to 1.000. Dipstick SG pads are unreliable in animals.

#328 · Laboratory Procedures

What urine SG range is isosthenuria, and what is concentrated in a cat?

Answer

Isosthenuria is 1.008 to 1.012. A cat concentrates above 1.035 and a dog above 1.030.

#329 · Laboratory Procedures

Which urine collection method is best for culture?

Answer

Cystocentesis, which avoids contamination from the lower tract and skin.

#330 · Laboratory Procedures

What is the urine sediment procedure?

Answer

Centrifuge about 5 mL for 5 minutes at 1,000 to 1,500 rpm, pour off most supernatant leaving 0.5 to 1 mL, resuspend and examine a drop under a coverslip.

#331 · Laboratory Procedures

Which urine crystals form in alkaline urine with a urease infection?

Answer

Struvite, with coffin-lid shapes.

#332 · Laboratory Procedures

Why is bilirubinuria more significant in a cat than a dog?

Answer

Any bilirubin in cat urine is abnormal. Small amounts can be normal in concentrated dog urine.

#333 · Laboratory Procedures

Which pad on a urine dipstick is unreliable in dogs and cats?

Answer

The leukocyte esterase pad, along with urobilinogen.

#334 · Laboratory Procedures

How does a fecal flotation solution work, and which suits Giardia?

Answer

The solution is denser than most eggs so they rise to the coverslip. Zinc sulfate (about 1.18) preserves Giardia cysts.

#335 · Laboratory Procedures

Describe a hookworm egg compared with a roundworm egg.

Answer

Hookworm eggs are oval and thin-shelled with 4 to 8 cells inside. Roundworm eggs are round with a thick pitted shell.

#336 · Laboratory Procedures

What does a heartworm antigen test detect?

Answer

Antigen from adult female worms. It can be negative in early infection or in a cat with few worms.

#337 · Laboratory Procedures

Why is a Wood's lamp negative result not a ringworm rule-out?

Answer

Only some Microsporum canis strains fluoresce apple green, so culture or microscopy is needed.

#338 · Laboratory Procedures

What is the best way to find Demodex mites?

Answer

A deep skin scrape until capillary bleeding.

#339 · Laboratory Procedures

What does MacConkey agar select for?

Answer

Gram-negative bacteria. Lactose fermenters grow pink.

#340 · Laboratory Procedures

What colours do gram-positive and gram-negative bacteria stain?

Answer

Gram-positive purple and gram-negative pink to red. Over-decolourizing makes gram-positive cells look pink.

#341 · Dentistry

What is the adult dog dental formula and total tooth count?

Answer

2 x (I 3/3, C 1/1, P 4/4, M 2/3) = 42 permanent teeth.

#342 · Dentistry

What is the adult cat dental formula and total tooth count?

Answer

2 x (I 3/3, C 1/1, P 3/2, M 1/1) = 30 permanent teeth.

#343 · Dentistry

How many deciduous teeth do puppies and kittens have?

Answer

Puppies have 28 and kittens have 26. Neither has deciduous molars.

#344 · Dentistry

By about what age should all permanent teeth be erupted in dogs and cats?

Answer

About 6 to 7 months of age.

#345 · Dentistry

What is a retained deciduous tooth and who treats it?

Answer

A baby tooth still present beside its erupted permanent tooth. The veterinarian extracts it.

#346 · Dentistry

Which teeth are the carnassials in dogs and cats?

Answer

The upper fourth premolar and the lower first molar.

#347 · Dentistry

How many roots do the upper fourth premolar and lower first molar have?

Answer

The upper fourth premolar has three roots and the lower first molar has two.

#348 · Dentistry

What is the hardest tissue in the body, and can it repair itself?

Answer

Enamel. It has no living cells and cannot repair itself.

#349 · Dentistry

What is the periodontium?

Answer

The supporting tissues of the tooth: gingiva, periodontal ligament, cementum and alveolar bone.

#350 · Dentistry

Define mesial and distal.

Answer

Mesial is toward the midline along the arch. Distal is away from the midline.

#351 · Dentistry

Which surface of a premolar faces the cheek?

Answer

The buccal surface.

#352 · Dentistry

In the Modified Triadan system, what does the first digit of 3 mean?

Answer

A permanent tooth in the left lower quadrant.

#353 · Dentistry

In the Modified Triadan system, which digits mean deciduous quadrants?

Answer

5 right upper, 6 left upper, 7 left lower, 8 right lower.

#354 · Dentistry

What Triadan number is the right upper carnassial of a dog?

Answer

108.

#355 · Dentistry

What Triadan number is the right lower canine of an adult dog?

Answer

404. The canine is always 04.

#356 · Dentistry

What Triadan number is the left upper canine of a puppy?

Answer

604.

#357 · Dentistry

What Triadan number is the left lower first molar?

Answer

309, the left lower carnassial.

#358 · Dentistry

How is plaque different from calculus?

Answer

Plaque is a soft film of bacteria and residue. Calculus is plaque that has mineralized, within days, and is rough.

#359 · Dentistry

What is the difference between gingivitis and periodontitis?

Answer

Gingivitis is gum inflammation only and is reversible. Periodontitis involves attachment loss of the PDL and bone and is not reversible.

#360 · Dentistry

How is periodontal disease staged?

Answer

By attachment loss: Stage 1 gingivitis only, 2 under 25 percent, 3 25 to 50 percent, 4 over 50 percent.

#361 · Dentistry

What does a pink or purple tooth after trauma suggest?

Answer

Bleeding into the pulp and probably a nonvital tooth. Take radiographs.

#362 · Dentistry

What is the most common oral cancer in cats?

Answer

Squamous cell carcinoma.

#363 · Dentistry

What is a base narrow canine?

Answer

A lower canine that angles inward and contacts the palate or upper gum.

#364 · Dentistry

What is the most important rule about polishing after scaling?

Answer

Always polish after scaling, because scaling scratches enamel. Never polish an unscaled tooth.

#365 · Dentistry

How long should a prophy cup stay on one tooth?

Answer

About 3 to 5 seconds with light pressure, because longer contact makes heat that can injure the pulp.

#366 · Dentistry

Why is a cuffed endotracheal tube required for dental procedures?

Answer

It protects the airway from aspiration of water, plaque and debris.

#367 · Dentistry

A 20 kg dog is on a 5 mL/kg/hr fluid rate for a dental. What is the pump rate?

Answer

20 x 5 = 100 mL/hr.

#368 · Dentistry

A 5 kg cat is limited to bupivacaine 1 mg/kg. How many mL of 0.5 percent is the maximum?

Answer

5 mg total. 0.5 percent is 5 mg/mL, so 1 mL.

#369 · Dentistry

How do you make 200 mL of 0.12 percent chlorhexidine from a 2 percent stock?

Answer

12 mL of stock plus 188 mL water, since 2 x 12 = 0.12 x 200.

#370 · Dentistry

Which instrument is used below the gum line, a sickle scaler or a curette?

Answer

The curette, which has a rounded toe. The pointed sickle is for above the gum only.

#371 · Dentistry

How do you test if a hand instrument is sharp?

Answer

Use a plastic test stick. A sharp edge bites in, a dull edge slides.

#372 · Dentistry

What must always be running when using an ultrasonic scaler or high-speed handpiece?

Answer

Water, to cool the tip and prevent heat damage to the pulp.

#373 · Dentistry

When is the bisecting angle technique used?

Answer

For upper teeth and lower incisors and canines, where the receptor cannot lie parallel. Aim the beam at a right angle to the line bisecting the tooth and receptor angle.

#374 · Dentistry

What error results if the beam is too close to perpendicular to the receptor?

Answer

Foreshortening, so the tooth looks too short.

#375 · Dentistry

What does a tooth with a wider pulp chamber than its opposite partner suggest?

Answer

It is probably nonvital, because dead pulp stops laying down dentin.

#376 · Emergency Medicine and Critical Care

What are the four main types of shock?

Answer

Hypovolemic, cardiogenic, distributive and obstructive.

#377 · Emergency Medicine and Critical Care

What signs mark compensatory shock in a dog?

Answer

Tachycardia, bright red or injected membranes, CRT under 1 second and bounding pulses.

#378 · Emergency Medicine and Critical Care

How does a cat in shock usually differ from a dog?

Answer

Cats are often bradycardic and hypothermic with pale membranes and weak pulses, so do not wait for tachycardia.

#379 · Emergency Medicine and Critical Care

What is hypotension by Doppler in a dog or cat?

Answer

A systolic pressure below 90 mmHg or a mean arterial pressure below 60 mmHg.

#380 · Emergency Medicine and Critical Care

What is the normal urine output in a hospitalized patient?

Answer

1 to 2 mL/kg/hr. Under 1 mL/kg/hr is oliguria and is reported.

#381 · Emergency Medicine and Critical Care

What do you do first when a dyspneic cat arrives?

Answer

Give oxygen by flow-by and keep stress low. Do not restrain for radiographs until stable.

#382 · Emergency Medicine and Critical Care

What is the primary survey order?

Answer

Airway, breathing, circulation, then disability and exposure.

#383 · Emergency Medicine and Critical Care

What compression rate does CPR use in dogs and cats?

Answer

100 to 120 compressions per minute.

#384 · Emergency Medicine and Critical Care

How deep are chest compressions in CPR?

Answer

One third to one half of the chest width, with full recoil.

#385 · Emergency Medicine and Critical Care

How often should CPR compressors rotate?

Answer

Every 2 minutes, to keep compressions effective.

#386 · Emergency Medicine and Critical Care

How is ventilation delivered once the patient is intubated during CPR?

Answer

About 10 breaths per minute, about 1 second each, without pausing compressions.

#387 · Emergency Medicine and Critical Care

What compression to breath ratio is used without an airway?

Answer

30 compressions to 2 breaths.

#388 · Emergency Medicine and Critical Care

Which ECG rhythms are shockable during CPR?

Answer

Ventricular fibrillation and pulseless ventricular tachycardia.

#389 · Emergency Medicine and Critical Care

What is the low dose epinephrine dose in CPR?

Answer

0.01 mg/kg IV, repeated every 3 to 5 minutes.

#390 · Emergency Medicine and Critical Care

What is the dose of atropine for asystole or high vagal tone arrest?

Answer

0.04 mg/kg IV.

#391 · Emergency Medicine and Critical Care

What does a sudden rise in ETCO2 during CPR suggest?

Answer

Possible return of spontaneous circulation.

#392 · Emergency Medicine and Critical Care

What is the shock fluid dose for a dog and for a cat?

Answer

Dog up to 90 mL/kg in the first hour, cat up to 40 to 60 mL/kg.

#393 · Emergency Medicine and Critical Care

How is a shock bolus given?

Answer

One quarter to one third of the dose over 15 minutes, then reassess.

#394 · Emergency Medicine and Critical Care

Calculate: a 20 kg dog needs a quarter of a 90 mL/kg shock dose. How many mL?

Answer

90 x 20 = 1,800 mL, and one quarter is 450 mL, given over 15 minutes.

#395 · Emergency Medicine and Critical Care

Which fluid can be mixed with blood products?

Answer

Only 0.9 percent sodium chloride. Lactated Ringer's has calcium and can cause clotting.

#396 · Emergency Medicine and Critical Care

Why must a cat be blood typed before its first transfusion?

Answer

Cats have natural alloantibodies, and a type B cat given type A blood can have a fatal reaction.

#397 · Emergency Medicine and Critical Care

Which dog blood group makes a universal donor?

Answer

A DEA 1 negative dog.

#398 · Emergency Medicine and Critical Care

What is the starting rate for a blood transfusion?

Answer

0.25 mL/kg/hr for the first 15 to 30 minutes while watching for a reaction.

#399 · Emergency Medicine and Critical Care

Within how many hours should a unit of blood be finished?

Answer

4 hours.

#400 · Emergency Medicine and Critical Care

What do you do if a transfusion reaction occurs?

Answer

Stop the transfusion, keep IV access with saline and alert the veterinarian.

#401 · Emergency Medicine and Critical Care

Calculate: a 20 kg dog, PCV 15, goal 25, donor PCV 40. Whole blood volume?

Answer

20 x 90 x 10 / 40 = 450 mL.

#402 · Emergency Medicine and Critical Care

At what rectal temperature do you stop active cooling for heat stroke?

Answer

39.5 degrees C.

#403 · Emergency Medicine and Critical Care

Which toxin causes low blood sugar and liver failure in dogs?

Answer

Xylitol.

#404 · Emergency Medicine and Critical Care

What is the antidote for anticoagulant rodenticide toxicity?

Answer

Vitamin K1, given for weeks.

#405 · Emergency Medicine and Critical Care

When is inducing vomiting contraindicated?

Answer

After caustics, hydrocarbons or sharp objects, and in seizing, depressed or dyspneic patients.

#406 · Emergency Medicine and Critical Care

Why is hydrogen peroxide not used to induce vomiting in cats?

Answer

It is not safe for cats, so the veterinarian chooses another emetic.

#407 · Emergency Medicine and Critical Care

What ECG changes occur with hyperkalemia in an obstructed cat?

Answer

Bradycardia, tented T waves, flattened or absent P waves and a wide QRS.

#408 · Emergency Medicine and Critical Care

How does calcium gluconate help hyperkalemia?

Answer

It protects the heart but does not lower the potassium.

#409 · Emergency Medicine and Critical Care

Calculate RER for a 10 kg dog using the linear formula.

Answer

30 x 10 + 70 = 370 kcal/day.

#410 · Emergency Medicine and Critical Care

At what oxygen level and duration does oxygen toxicity become a risk?

Answer

More than 60 percent oxygen for more than 24 hours.

#411 · Pain Management and Analgesia

Name the four steps of the nociception pathway in order.

Answer

Transduction (stimulus becomes an electrical signal at the nociceptor), transmission (signal travels to the spinal cord and brain), modulation (signal is amplified or dampened in the spinal cord), and perception (the brain registers pain).

#412 · Pain Management and Analgesia

Which nerve fibres carry fast, sharp pain and which carry slow, dull, aching pain?

Answer

A-delta fibres carry fast, sharp, well-localized pain. C fibres carry slow, dull, burning or aching pain that is harder to localize.

#413 · Pain Management and Analgesia

What is central sensitization (wind-up) and why does it matter for treatment?

Answer

Repeated painful input makes spinal cord neurons hyperexcitable through NMDA receptor activation, so normal touch can hurt and pain is felt more strongly. Treating pain early and with several drug classes helps prevent it.

#414 · Pain Management and Analgesia

What is the difference between hyperalgesia and allodynia?

Answer

Hyperalgesia is an exaggerated response to a stimulus that is normally painful. Allodynia is pain caused by a stimulus that is not normally painful, such as light touch.

#415 · Pain Management and Analgesia

What is multimodal analgesia?

Answer

Using two or more drug classes or techniques that act at different points on the pain pathway, such as an opioid plus an NSAID plus a local block. It gives better relief with lower doses of each drug and fewer side effects.

#416 · Pain Management and Analgesia

What is pre-emptive analgesia?

Answer

Giving analgesics before the painful stimulus, such as before surgery. It reduces central sensitization and lowers the amount of analgesia needed afterward.

#417 · Pain Management and Analgesia

Does the absence of vocalizing mean an animal is not in pain?

Answer

No. Many animals, especially cats and prey species such as rabbits and birds, hide pain. Quiet, hiding, reduced appetite and reluctance to move can all be signs of significant pain.

#418 · Pain Management and Analgesia

List common behavioural signs of pain in a dog.

Answer

Whimpering or yelling, guarding or licking an area, restlessness or hiding, reluctance to move or rise, hunched posture, panting, decreased appetite, and sometimes aggression when touched.

#419 · Pain Management and Analgesia

List common signs of pain in a cat.

Answer

Hiding, hunched posture with eyes squinted, reduced grooming, decreased appetite, reluctance to jump, hissing or growling when approached, and sometimes sitting motionless at the back of the cage.

#420 · Pain Management and Analgesia

Which facial features does the Feline Grimace Scale score?

Answer

Ear position, orbital tightening, muzzle tension, whisker position and head position. Each is scored 0 to 2.

#421 · Pain Management and Analgesia

What score on the short form Glasgow Composite Measure Pain Scale suggests a dog needs analgesia?

Answer

A total of 6 or more out of 24 (or 5 or more out of 20 if mobility cannot be assessed) suggests analgesia is needed. The result is a guide and the veterinarian makes the final decision.

#422 · Pain Management and Analgesia

Why should pain scores be repeated and recorded?

Answer

A single score is a snapshot. Reassessing at regular intervals and after each drug shows whether the plan works, lets the veterinarian adjust it, and documents patient comfort in the medical record.

#423 · Pain Management and Analgesia

Name four full mu agonist opioids used in small animal practice.

Answer

Morphine, hydromorphone, fentanyl and methadone (oxymorphone is also a full mu agonist). They give strong analgesia for moderate to severe pain.

#424 · Pain Management and Analgesia

How does buprenorphine differ from full mu agonists?

Answer

It is a partial mu agonist with high receptor affinity. It gives mild to moderate analgesia, lasts about 6 to 8 hours, and is hard to reverse with naloxone. It is well absorbed across the oral mucosa in cats.

#425 · Pain Management and Analgesia

How does butorphanol differ from morphine in analgesic strength and duration?

Answer

Butorphanol is a kappa agonist and mu antagonist. It gives only mild, short-lived analgesia (often 1 to 2 hours) with good sedation and cough suppression, so it is not suited to severe pain.

#426 · Pain Management and Analgesia

What are common side effects of opioids?

Answer

Sedation, respiratory depression, bradycardia, panting, vomiting (especially morphine in dogs), decreased gut motility, urinary retention and, in some animals, dysphoria or excitement.

#427 · Pain Management and Analgesia

How is opioid overdose or marked respiratory depression treated?

Answer

Support ventilation and oxygen, then give naloxone as ordered by the veterinarian. Naloxone lasts a shorter time than most opioids, so watch for the animal slipping back into depression and re-dosing may be needed.

#428 · Pain Management and Analgesia

Why is morphine given slowly or by IM injection rather than as a rapid IV bolus?

Answer

Morphine can release histamine, and a fast IV bolus may cause hypotension and itching. A slow injection or the IM route reduces this.

#429 · Pain Management and Analgesia

Calculate: a 20 kg dog is ordered morphine 0.5 mg/kg IM. The vial is 15 mg/mL. How many mL?

Answer

20 kg x 0.5 mg/kg = 10 mg. 10 mg / 15 mg/mL = 0.67 mL.

#430 · Pain Management and Analgesia

Calculate: a fentanyl CRI of 5 mcg/kg/hr is ordered for a 10 kg dog. Fentanyl is 50 mcg/mL. What pump rate in mL/hr?

Answer

10 kg x 5 mcg/kg/hr = 50 mcg/hr. 50 mcg/hr / 50 mcg/mL = 1 mL/hr.

#431 · Pain Management and Analgesia

What is the usual delay before a fentanyl patch gives effective analgesia?

Answer

Roughly 12 to 24 hours, so another analgesic is needed in the meantime. Patches must be kept away from heat, and the used patch is a controlled-drug hazard to children and other pets.

#432 · Pain Management and Analgesia

Under which Canadian law are opioids, ketamine and benzodiazepines controlled, and what record is required?

Answer

The Controlled Drugs and Substances Act. The clinic must keep secure storage and a controlled drug log recording each drug received, used and wasted, with regular counts.

#433 · Pain Management and Analgesia

What is the mechanism of NSAIDs?

Answer

They inhibit cyclooxygenase (COX) enzymes, reducing prostaglandins that cause inflammation, pain and fever. Blocking protective prostaglandins is also what causes the GI and kidney side effects.

#434 · Pain Management and Analgesia

List the main adverse effects of NSAIDs in dogs and cats.

Answer

Vomiting, diarrhea, GI ulceration and bleeding, kidney injury (especially with dehydration or hypotension) and liver injury. Clients should stop the drug and call if the pet vomits, has black stool or stops eating.

#435 · Pain Management and Analgesia

Why must NSAIDs not be given together with corticosteroids?

Answer

The combination greatly raises the risk of GI ulceration and perforation. A washout period set by the veterinarian is needed when switching between the two.

#436 · Pain Management and Analgesia

Why must owners never give human ibuprofen, naproxen or acetaminophen to pets?

Answer

Ibuprofen and naproxen cause serious ulcers and kidney failure in dogs and cats. Acetaminophen is especially toxic to cats, damaging red blood cells and the liver.

#437 · Pain Management and Analgesia

Calculate: a 30 kg dog is ordered meloxicam 0.1 mg/kg. The oral suspension is 1.5 mg/mL. How many mL?

Answer

30 kg x 0.1 mg/kg = 3 mg. 3 mg / 1.5 mg/mL = 2 mL.

#438 · Pain Management and Analgesia

Compare lidocaine and bupivacaine as local anesthetics.

Answer

Lidocaine has fast onset and a short duration (about 1 to 2 hours). Bupivacaine is slower to start but lasts roughly 4 to 8 hours, and is more cardiotoxic if injected intravenously by mistake.

#439 · Pain Management and Analgesia

What safety step is done before injecting a local anesthetic for a nerve block?

Answer

Aspirate to confirm the needle is not in a vessel, and keep to the maximum dose. Cats tolerate lower doses than dogs, so the calculation must use the patient's species and weight.

#440 · Pain Management and Analgesia

What are the effects of alpha-2 agonists such as dexmedetomidine, and how are they reversed?

Answer

Sedation and analgesia, with bradycardia and peripheral vasoconstriction (pale mucous membranes). Atipamezole reverses them, which also reverses their analgesia.

#441 · Pain Management and Analgesia

What conditions is gabapentin commonly used for?

Answer

Chronic and neuropathic pain, often as part of multimodal therapy. Its main side effect is sedation, and it is also used before visits to reduce anxiety in cats.

#442 · Pain Management and Analgesia

What is the role of low-dose ketamine in pain control?

Answer

Ketamine blocks NMDA receptors, which reduces wind-up. It is given as a low-dose CRI with opioids for severe or chronic pain. It is a controlled drug in Canada.

#443 · Pain Management and Analgesia

What non-drug measures support pain management in hospitalized patients?

Answer

Soft, clean, dry bedding, a quiet low-stress area, gentle handling, cold packs for the first days after injury or surgery then warmth, passive range of motion, turning recumbent patients, and short, calm walks when allowed.

#444 · Pain Management and Analgesia

What do you do first if a postoperative patient scores high on the pain scale?

Answer

Check that the patient is stable, tell the veterinarian right away with the score and observations, give the ordered rescue analgesic, and reassess after the drug has had time to act.

#445 · Pain Management and Analgesia

Which clinical signs commonly indicate pain in a horse with colic?

Answer

Pawing, flank watching, rolling, repeated lying down and getting up, stretching out, sweating and an elevated heart rate. Heart rate above about 60 beats per minute suggests severe pain.

#446 · Diagnostic Imaging

What does kVp control on a radiograph?

Answer

Beam energy and penetration, so it controls contrast. Higher kVp gives a greyer, lower contrast image.

#447 · Diagnostic Imaging

What does mAs control on a radiograph?

Answer

The quantity of x-rays made, so it controls density. More mAs gives a darker image.

#448 · Diagnostic Imaging

A radiograph is too light. Which setting do you raise?

Answer

Raise the mAs (or the kVp). Too little exposure makes an underexposed, light image.

#449 · Diagnostic Imaging

What is the 15 percent rule?

Answer

Raising kVp by 15 percent has about the same effect on density as doubling mAs. Lowering kVp by 15 percent matches halving mAs.

#450 · Diagnostic Imaging

You expose at 70 kVp and 10 mAs, then raise kVp by 15 percent. What mAs keeps the same density?

Answer

New kVp is 70 x 1.15 = 80.5. Halve the mAs to 5.

#451 · Diagnostic Imaging

State the inverse square law for x-rays.

Answer

Beam intensity is inversely proportional to the square of the distance. Doubling the distance gives one quarter of the intensity.

#452 · Diagnostic Imaging

10 mAs works at 100 cm. What mAs is needed at 80 cm?

Answer

mAs2 = 10 x (80 / 100) squared = 10 x 0.64 = 6.4 mAs. Closer means less mAs.

#453 · Diagnostic Imaging

How do you pick settings from a technique chart?

Answer

Measure the thickest part of the area in cm with calipers and read the kVp and mAs for that thickness on the chart for your machine.

#454 · Diagnostic Imaging

When is a grid used?

Answer

When the part is thicker than about 10 cm. It absorbs scatter and sharpens the image, but you must raise the technique.

#455 · Diagnostic Imaging

What does grid cutoff look like and what causes it?

Answer

A loss of density in a band or at the edges. The grid is off centre, tilted, upside down or at the wrong distance.

#456 · Diagnostic Imaging

What does collimation do?

Answer

It narrows the beam to the area of interest, lowering dose and scatter and improving image quality.

#457 · Diagnostic Imaging

What is the heel effect and how do you use it?

Answer

The beam is weaker on the anode side. Put the thicker end of the part toward the cathode side.

#458 · Diagnostic Imaging

What does ALARA stand for?

Answer

As low as reasonably achievable. Use time, distance and shielding to keep exposure low.

#459 · Diagnostic Imaging

How do you keep a dog still for a radiograph without holding it?

Answer

Sedation or anesthesia plus sandbags, tape, troughs and positioning aids. Nobody holds the patient routinely.

#460 · Diagnostic Imaging

Do lead gloves allow a hand in the primary beam?

Answer

No. Gloves stop scatter only. No part of the body should ever be in the primary beam.

#461 · Diagnostic Imaging

Where do you wear a dosimeter badge?

Answer

At collar level, outside the lead apron, facing the tube. Keep a control badge away from radiation.

#462 · Diagnostic Imaging

How should lead aprons be stored?

Answer

Hung on a rack or laid flat, never folded, because folding cracks the lead. Inspect them on a schedule.

#463 · Diagnostic Imaging

Name the five radiographic opacities from darkest to brightest.

Answer

Gas, fat, soft tissue and fluid, bone, metal.

#464 · Diagnostic Imaging

What does a right lateral radiograph mean?

Answer

The animal lies on its right side, with the right side closest to the detector.

#465 · Diagnostic Imaging

What is the difference between a VD and a DV view?

Answer

VD: the beam enters the belly side, so the animal is on its back. DV: the beam enters at the back, so the animal is on its belly.

#466 · Diagnostic Imaging

When in the breathing cycle is a chest radiograph taken?

Answer

At peak inspiration, when the lungs are full of air.

#467 · Diagnostic Imaging

When is an abdominal radiograph taken?

Answer

At the end of expiration.

#468 · Diagnostic Imaging

A cat's chest radiograph is blurred. What is the fix?

Answer

Shorten the exposure time by using a higher mA, and sedate if needed. The blur is motion.

#469 · Diagnostic Imaging

What should every radiograph carry for labelling?

Answer

Patient name, owner or file number, date, clinic and a left or right marker placed in the field at the time of exposure.

#470 · Diagnostic Imaging

How many views of the chest are standard?

Answer

Three: right and left laterals plus a VD or DV.

#471 · Diagnostic Imaging

What does an anechoic structure on ultrasound mean?

Answer

No echoes, shown black. It is fluid, such as urine or bile.

#472 · Diagnostic Imaging

What frequency probe suits a large deep dog versus a cat?

Answer

Lower frequency (3 to 5 MHz) for deep, large patients. Higher frequency (7.5 to 12 MHz) for small patients and shallow organs.

#473 · Diagnostic Imaging

Why clip the hair and use gel for ultrasound?

Answer

Air trapped in hair or between probe and skin blocks sound and ruins the image.

#474 · Diagnostic Imaging

Which contrast agent is used if gut perforation is suspected?

Answer

A water soluble iodinated agent. Barium leaking into the abdomen can cause peritonitis.

#475 · Diagnostic Imaging

What risk comes with iodinated contrast and how do you reduce it?

Answer

It is cleared by the kidneys, so dehydrated or renal patients are at risk. Hydrate the patient and monitor for reactions.

#476 · Communication and Veterinary Professional Support Services

What does the VTNE passing score of 425 represent?

Answer

It is a scaled score on a 200 to 800 scale, not a percentage. The number of correct answers does not convert directly to a percent.

#477 · Communication and Veterinary Professional Support Services

How many questions are on the VTNE and how long do candidates have?

Answer

150 scored plus 20 unscored multiple choice questions, in 3 hours. Candidates cannot tell which are unscored.

#478 · Communication and Veterinary Professional Support Services

Who regulates registration of veterinary technicians in Canada?

Answer

Each province, through its own association or college. The candidate must confirm current steps, fees and retake limits with their own regulator.

#479 · Communication and Veterinary Professional Support Services

Which Canadian provinces does the AAVSB list as regulated RVT provinces (Aug 2025 list)?

Answer

Alberta, Manitoba, Nova Scotia, Ontario and Saskatchewan. Lists change, so confirm with your regulator.

#480 · Communication and Veterinary Professional Support Services

What is the teach-back method?

Answer

Ask the client to explain the instructions back in their own words. It tests how well you explained, not the client.

#481 · Communication and Veterinary Professional Support Services

What are the four parts of a SOAP entry?

Answer

Subjective (client report), Objective (measured findings), Assessment (problem or diagnosis by the veterinarian), Plan (treatments, diagnostics, instructions).

#482 · Communication and Veterinary Professional Support Services

How do you correct an error in a paper medical record?

Answer

Draw one line through it so it stays legible, write error with the date and your initials, and add the correct entry. Never erase or white out.

#483 · Communication and Veterinary Professional Support Services

What is signalment?

Answer

The basic description of the patient: species, breed, age, sex and reproductive status, and often color and weight.

#484 · Communication and Veterinary Professional Support Services

A client asks a technician whether the lump is cancer. What is the correct response?

Answer

Explain that the veterinarian will go over the results, and arrange that conversation. Technicians do not diagnose or give a prognosis.

#485 · Communication and Veterinary Professional Support Services

What should you do first when a client is angry?

Answer

Stay calm, listen without interrupting, and acknowledge the feeling. Then state what you can do and involve the veterinarian or manager if needed.

#486 · Communication and Veterinary Professional Support Services

Who may receive patient and owner information without extra consent?

Answer

The owner, the care team, people the owner has authorized, and where the law requires it. A neighbour or non-owner relative is not entitled to details.

#487 · Communication and Veterinary Professional Support Services

What is closed-loop communication in an emergency?

Answer

The receiver repeats the order back, carries it out, then confirms it is done, for example giving the drug and name and dose. This prevents dosing errors.

#488 · Communication and Veterinary Professional Support Services

What must be checked before a euthanasia begins?

Answer

The signed consent from the owner or authorized agent, and that the patient identification matches the form.

#489 · Communication and Veterinary Professional Support Services

Which body signs does the veterinarian use to confirm death after euthanasia?

Answer

Absence of heartbeat and breathing and loss of the corneal reflex.

#490 · Communication and Veterinary Professional Support Services

What normal reactions may follow euthanasia that clients should be told about?

Answer

Muscle twitches, a deep breath after the heart stops, urination or defecation and open eyes. These are not signs of pain.

#491 · Communication and Veterinary Professional Support Services

Which drug class is commonly used for euthanasia and how is it stored?

Answer

A barbiturate such as pentobarbital, given intravenously by the veterinarian. It is a controlled substance, locked and logged.

#492 · Communication and Veterinary Professional Support Services

What should a controlled drug log entry record?

Answer

Date, patient, drug, amount used, amount wasted, remaining balance and who gave it. Stock is counted against the log and discrepancies are reported to the veterinarian at once.

#493 · Communication and Veterinary Professional Support Services

Which Canadian law controls opioids, ketamine, benzodiazepines and barbiturates?

Answer

The Controlled Drugs and Substances Act (CDSA).

#494 · Communication and Veterinary Professional Support Services

Name injuries that suggest animal abuse rather than accident.

Answer

Injuries that do not match the history, fractures at different stages of healing, burns, rope or wire marks and patterned bruising.

#495 · Communication and Veterinary Professional Support Services

What should a technician do when abuse is suspected?

Answer

Document objectively, tell the veterinarian right away, and do not accuse or confront the owner. The veterinarian decides what is reported.

#496 · Communication and Veterinary Professional Support Services

What is the first step in crisis intervention?

Answer

Make sure everyone is safe, including yourself. Then stay calm, listen, and bring in the veterinarian or manager if needed.

#497 · Communication and Veterinary Professional Support Services

What is compassion fatigue?

Answer

A gradual loss of empathy and energy after prolonged exposure to suffering. Support includes breaks, peer support, debriefing and professional help.

#498 · Communication and Veterinary Professional Support Services

A technician realizes they gave the wrong dose. What should they do?

Answer

Tell the veterinarian immediately so the patient can be treated, then document the event factually. Hiding errors is the worst choice.

#499 · Communication and Veterinary Professional Support Services

What does informed consent require?

Answer

The owner understands the procedure, risks, benefits, alternatives and cost, and agrees voluntarily. The veterinarian explains the medical details.

#500 · Communication and Veterinary Professional Support Services

Why should pregnant clients avoid cleaning cat litter boxes?

Answer

Cat feces can carry Toxoplasma gondii, a zoonotic parasite that can harm a developing fetus.